

Osamah I. Alqaisi

+966-550990851 | OALQAISI@UT.EDU.SA | [LinkedIn](#) | [Google Scholar](#) | [Website](#)

Assistant Professor of Computer Science at the University of Tabuk. Research interests include Internet of Things (IoT), Machine Learning, Computer Vision, Edge Computing, and Container Technologies. Experienced in teaching, research, curriculum development, and supervising undergraduate projects, with publications in leading conferences and journals.

Education

- PhD, Computer Science – University of Texas at San Antonio, USA - (GPA: 4/4) - Aug 2019 to May 2024
 - Master, Computer Science – University of Texas at San Antonio, USA - (GPA: 4/4) - Aug 2019 to Dec 2023
 - Master, Computer Science – Western Michigan University, USA - Aug 2014 to Jul 2016
 - B.Sc., Computer Science – Taibah University, KSA - Aug 2010 to May 2014
-

Experience

- Assistant Professor - CS Department - University of Tabuk, KSA (Dec 2024 – Present)
 - Lecturer - CS Department - University of Tabuk, KSA (Jan 2023 – Nov 2024)
 - Teaching Assistant & Grader - CS Department - University of Texas at San Antonio, TX USA (Jan 2021 – May 2024)
 - Tutor - College of Engineering and Applied Sciences - Western Michigan University - MI, USA (Jun 2014 – Jun 2016)
 - Teaching Assistant - CS Department - University of Tabuk, KSA (Jan 2012 – Dec 2022)
 - Data Entry – Seasonal - Saud Telecom Company (STC) – Riyadh, KSA (Jun 2007 – Sep 2007)
-

Professional Service

- Chair, Graduation Projects Committee - CS Department - University of Tabuk
 - Member, Conferences and Forums Committee - College of Computer & Information Technology - University of Tabuk
 - Member, Committee for Developing Programing, AI & ML Courses - CS Department - University of Tabuk
 - Member, Computer Science Department Council - CS Department - University of Tabuk
 - Member, Institutional Reputation Committee for Colleges - University of Tabuk
-

Publications

- Nassef, L., Alghamdi, M.I., Chaabane, S.B., Abbas, Q., Alawad, W.M., Albalawi, O.H., Alqaisi, O.I., & Fakieh, B. "Lightweight and Energy-Aware Intrusion Detection for Industrial IoT Using TinyML and Edge AI". Sci Rep. 2026 Jun 15;16(1):18524. doi: 10.1038/s41598-026-50690-0.
- S. Kaiser, O. Alqaisi, A. Ş. Tosun and T. Korkmaz, "Benchmarking Container Orchestrations for ARM-Powered Edge Computing," 2025 IEEE 22nd Consumer Communications & Networking Conference (CCNC), Las Vegas, NV, USA, 2025, pp. 1-9, doi: 10.1109/CCNC54725.2025.10976159.
- O. I. Alqaisi, A. Ş. Tosun and T. Korkmaz, "The Impact of GPU on Containerized Computer Vision Applications on Edge Nodes," 2024 IEEE International Conference on Edge Computing and Communications (EDGE), Shenzhen, China, 2024, pp. 74-83, doi: 10.1109/EDGE62653.2024.00019.
- O. I. Alqaisi, A. Ş. Tosun and T. Korkmaz, "Performance Analysis of Container Technologies for Computer Vision Applications on Edge Devices," in IEEE Access, doi: 10.1109/ACCESS.2024.3376570.
- O. I. Alqaisi, A. Şaman Tosun and T. Korkmaz, "Containerized Computer Vision Applications on Edge Devices," 2023 IEEE International Conference on Edge Computing and Communications (EDGE), Chicago, IL, USA, 2023, pp. 1-11, doi: 10.1109/EDGE60047.2023.00014.
- O. I. Alqaisi, M. S. Haq and A. S. Tosun, "Security of Containerized Computer Vision Applications," 2022 2nd International Conference on Computing and Information Technology (ICCIT), 2022, pp. 115-120, doi: 10.1109/ICCIT52419.2022.9711600.

Awards & Honors

- Excellence in E-Learning Award – Teaching Excellence Track, University of Tabuk, 2025.
- Certificate of Appreciation – Supervision of Student Projects for the 2025 National Science Olympiad, General Administration of Education, Tabuk Region (Educational Affairs – Talent Development Department), 2025.

Certifications and Workshops

- Faculty Workshop: AI for Everyone - atomcamp Arabia x MCIT - July 2025
- Creative evaluation: strategies and practical applications (3 Training hours) - University of Tabuk - Oct 2024
- Effective Time Management and Communication Strategies (New Faculty Orientation 1446 AH) (3 Training hours) - University of Tabuk - Sep 2024
- Modern Methods of Teaching: Tools and Strategies for Achieving Academic Excellence (New Faculty Orientation 1446 AH) (3 Training hours) - University of Tabuk - Sep 2024
- Data Management and Governance System at the University of Tabuk Training - University of Tabuk - Oct 2023
- Explanation of the executive regulations for academic promotions at the University of Tabuk (2 Training hours) - University of Tabuk - Oct 2023
- Anomaly-Based Intrusion Detection System in IoT for Healthcare Using Machine Learning Workshops - University of Tabuk - Oct 2023
- Introduction to Bioinformatics and next-generation sequencing Workshops - University of Tabuk - Jan 2023
- Educational Game Production Workshops - University of Tabuk - 2023
- Fundamentals of Deep Learning – Nvidia – Jan 2022.
- Standards of Conduct Training at UTSA – University of Texas at San Antonio - May 2021.
- Return to Campus Training for All UTSA Community Members – University of Texas at San Antonio - May 2021.
- Constructing Achievement Tests - University of Tabuk - Oct 2016.
- Dynamic of Innovative Engineering, Management, Leadership and Business Strategies - Western Michigan University - Feb 2016.
- Data Analytics - Western Michigan University - Jan 2014.

Skills

Technical Skills:

- Programming: C, C++, C#, Java, Python
- AI & Data Analysis: TensorFlow, PyTorch, Keras, Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib, Prompt Engineering
- Network & Cloud: MQTT, CoAP protocols, Container Orchestration, Docker
- Web: HTML, JavaScript, CSS, UX Design, CMS
- Database Management: MySQL
- Version Control: Git, GitHub/GitLab

Managerial & Soft Skills:

- Time Management
- Teamwork & Collaboration
- Adaptability
- Decision Making
- Problem Solving
- Critical & Creative Thinking
- Effective Communication
- Project Planning & Organization
- Emotional Intelligence